HemoLink AI

See a blood shortage coming — and reach the donors most likely to come.

DSH Hacks V2 · AI × Healthcare

Live demo: (link added after deploy)
Code: (link added after deploy)
Demo login: donor@demo.com · Demo123!

1 The problem

- Only **2.3% of Malaysians** donate blood; the National Blood Centre (PDN) is aiming for 3%.
- Klang Valley hospitals need **600–700 bags a day**, and red cells keep only 35–42 days (platelets 5), so collection can never pause.
- Shortages are noticed late, so blood banks send mass appeals that tire donors out. Many donors don't know if they're eligible and are turned away on the day.

2.3%	600–700	≈50%
donation rate vs 3% target	bags a day, Klang Valley	of Malaysians are group O

2 Our solution

One web app (it also installs on phones) that links the three people a blood supply depends on. AI helps each of them decide earlier and better:

Donor
Finds nearby drives, checks eligibility, books, answers urgent requests, understands their blood test.

Drive organizer
Clubs, companies, NGOs: run drives, check donors in, plan the next one.

Blood bank
Sees the 7-day shortage outlook and alerts just the right donors.

3 Key features

Shortage outlook AI

Forecasts donations and use for each of the 8 blood types and projects stock 7 days ahead. It warns *which* type will fall below the safe level, and *why*.

Smart donor matching AI

Ranks compatible, eligible donors by distance, past replies and recent alerts, then alerts only as many as needed (units × 1.5), not everyone.

Eligibility check AI

PDN's real rules decide: age 17–65, a 3-month gap for men and 4 for women, and 15 screening questions. The AI only explains the result, in English or Malay.

HemoScan AI

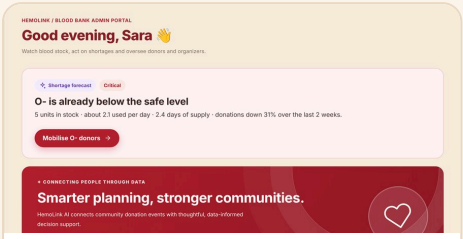
Reads a blood test PDF or photo. The donor confirms every value, then clear rules flag out-of-range patterns and trends. A guide to bring to the doctor, never a diagnosis.

Campaign map & planner

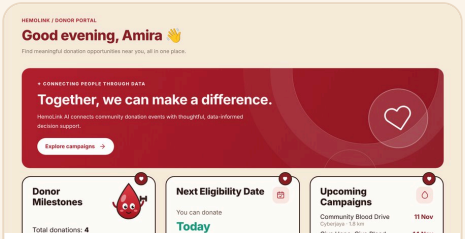
GPS distances to drives, organizers pin venues on a map, and the planner suggests the area and weekday with the most eligible donors.

Alerts that reach people

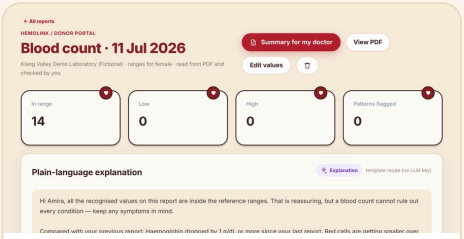
Alerts in the app, by email, SMS and phone push, with reminders the day before. The whole app works in English and Bahasa Malaysia.



Blood bank: O− flagged below the safe level → mobilise donors



Donor: milestones, next eligible date, drives nearby



HemoScan: values checked by the donor, explained in plain language

4 How the AI was built and tested

- **Real data:** Ministry of Health daily donations, 2006–2026, for 14 regions (the latest 3 years are used for training). 70 region × blood-group series, each choosing the best of 3 methods by rolling cross-validation. **27% lower error** than a simple "same day last week" forecast.
- **Donor return model:** predicts whether a donor will come back. AUC 0.85 (1.0 is perfect, 0.5 is guessing), checked 5 ways, and well calibrated (error 0.145 vs 0.213 for a basic guess).
- **Fairness check:** compares match rates across age groups and areas. Donors aged 45+ are picked a little less often (0.75–0.79 of the top group's rate). Age isn't scored; distance and donation history explain most of the gap, and admins can see this.
- **Safety:** rules make every decision and the language model only explains. Medical questions are filtered before they reach the AI, and every answer has a fallback that works without AI.

5 Trust, impact and next steps

- **Privacy (PDPA 2010):** consent is recorded, donors can download or delete their data, blood test files are encrypted, and organizers and admins can never see health data.
- **Impact:** fewer and better-targeted appeals, fewer donors turned away on the day, and shortages spotted days earlier.
- **Next:** pilot with a campus drive, connect real blood bank stock, add WhatsApp alerts, Chinese and Tamil.

Built with

Python · FastAPI | scikit-learn | SQLite | Gemini / Claude | Leaflet + OSM | Web push · PWA | 54 automated tests

Student prototype — not a medical device; screening staff make every decision. Blood stock, centres and most donors in the demo are simulated. Data: Ministry of Health Malaysia / National Blood Centre daily donations (CC BY 4.0); eligibility rules from PDN's donor registration form and FAQ (checked Nov 2026); statistics from PDN's Deputy Director via Bernama, 16 Jun 2026. Maps © OpenStreetMap contributors. Code licensed AGPL-3.0.